Liquid Shear → Liquid Based Homogenization

1. Cell distruption technique for small volumes and cultured cells.

2. Cells are lysed by forcing cells or tissue suspension through a narrow space.
∴ shearing the cell membrane

Types of Homogenizer :

(A) Dounce Homogenizer : Consist of round glass manually driven into pestel.

- Influenced by number and speed of strokes.

- Available in variety of sizes to accomodate different volumes.

100

(B) Potter Elvehjem Homogenizer:

Consist of manual or mechanically driven PFTE pestle shaped to fit around a conical or rounded vessel.

(C) French Press: Piston which is used to apply high amount of pressure on a volume of 40 – 250 ml. forcing it through a tiny hole

- 2 passes are required for effective lysis.

Stress developed in homogenizer (P_s) dynamic Pressure and is independent of density of fluid (ρ) and jet velocity (v)

1. $P_s = \frac{1}{2} \rho v^2$

2. $v = \frac{Q}{A}$

3. $\ln\left(\frac{C_{max}}{C_{max} - C}\right) = -kN$

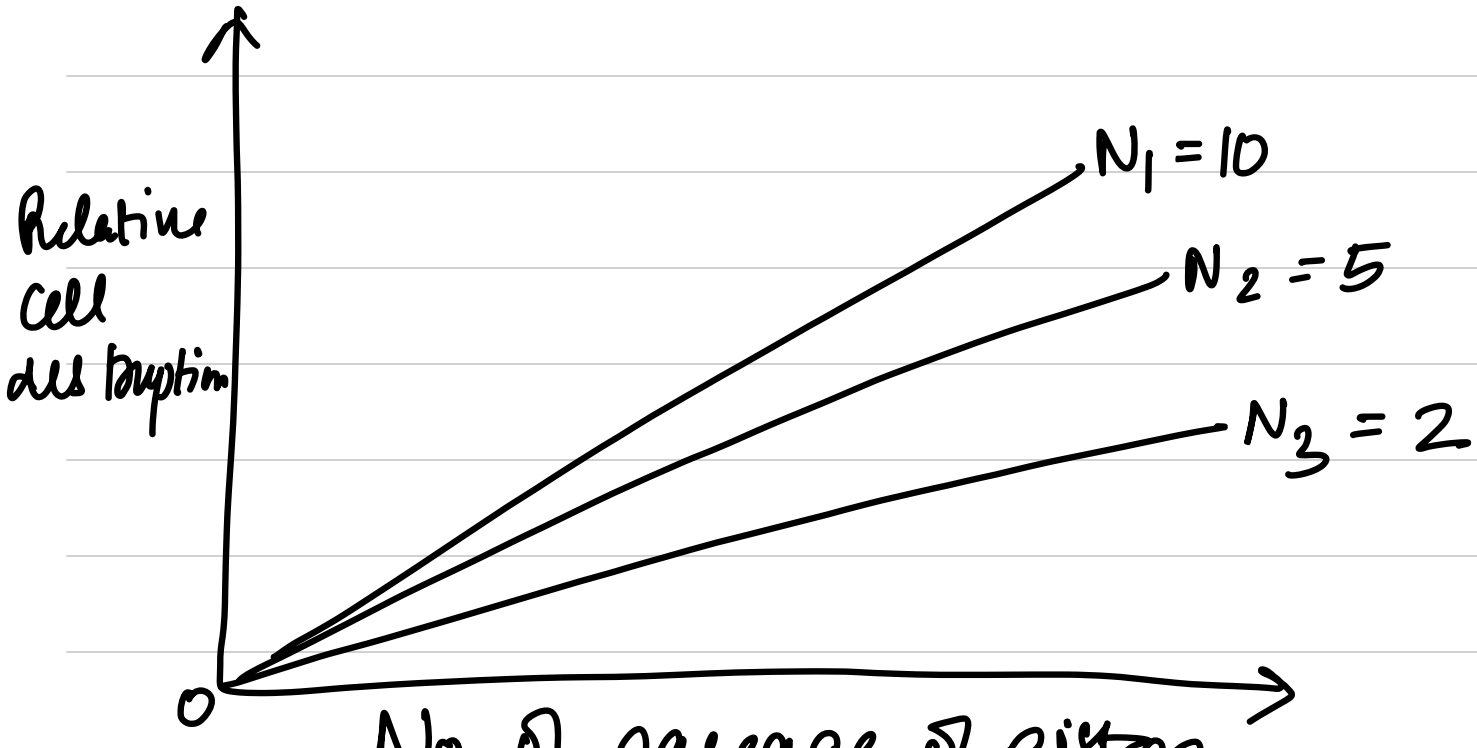
k → disruption rate const.

N → No. of passes through homogenizer

C_{max} = maximum # cell disruptions

$k = k' P_s^{n}$ → n → empirical constant.

$$\therefore \boxed{\ln\left(\frac{C_{max}}{C_{max} - C}\right) = k' P_s^{n} N}$$



No. of passage of piston

Higher Pressure & Cell Homogenization faster

Q2 Citric Acid

Citric Acid is a primary metabolite and an extracellular product. It is produced by micro-organisms and is dissolved in the broth. Steps for achieving pure citric acid are:

1. Cell Removal
   - Solid-Liquid Filteration / Centrifugation
2. Primary Isolation
3. Product Enrichment [Purification]
4. Final Isolation

Citric acid is a primary metabolite and an **extracellular product**, meaning it is

secreted by the microorganisms directly into the surrounding fermentation broth. Because it is extracellular, its purification bypasses the need for cell disruption, which is only required for products trapped inside the cells.

Based on the generic scheme for downstream processing of extracellular products, the major steps for purifying citric acid are:

1. Cell Removal (Solid-Liquid Separation)

The very first step is to separate the microbial cells and suspended biomass from the liquid broth. Because the citric acid is already dissolved in the liquid, solid-liquid separation techniques such as **filtration** or **centrifugation** are directly applied to remove the solids and recover the product-rich, cell-free liquor.

2. Primary Isolation

Once the cells are removed, the dilute cell-free liquor undergoes primary isolation to concentrate the product and remove contaminants that have properties widely different from citric acid. Typical unit operations at this stage include extraction, adsorption, or precipitation.

3. Product Enrichment (Purification)

The concentrated product stream is further processed to remove remaining impurities that might be chemically or physically similar to the citric acid. This stage typically utilizes high-resolution techniques such as chromatography or fractional precipitation.

4. Final Isolation (Polishing)

The last step is designed to remove remaining liquids and convert the purified citric acid into its final, stable form. For many biochemicals, this is achieved through **crystallization** and drying.

Would you like me to explain the different types of industrial filtration or centrifugation equipment that are typically used for that initial cell removal step?

Question-Define RCF value and write its significance. Calculate RCF for a tubular centrifuge with radius 12.7 cm, if the centrifuge is to be operated at 5000 rpm.

answer: **Definition and Significance of RCF**

Relative Centrifugal Force (RCF), denoted by Z and commonly described as "g-force," is the ratio of centrifugal force (F_c) to gravitational force (F_g).

The significance of RCF is that it measures the actual separating power of a centrifuge relative to Earth's gravity. It tells scientists exactly how strong the centrifuge is compared to natural gravity. For example, an RCF of 10,000 g means that the force exerted on the particles is 10,000 times stronger than gravity.

Because it normalizes the centrifugal force regardless of the machine's specific dimensions, RCF is the standard metric used to describe a centrifuge's power.

Calculation of RCF

The sources provide the simplified formula for RCF as:

$Z = 0.001119 \cdot r \cdot n^2$

Based on the parameters defined in the text, the rotor radius (r) must be in meters and the rotational speed (n) must be in rpm.

Given your values:

- Radius (r) = 12.7 cm = **0.127 m**
- Rotational speed (n) = **5000 rpm**

Plugging these into the equation:

- $Z = 0.001119 \times 0.127 \times (5000)^2$
- $Z = 0.001119 \times 0.127 \times 25,000,000$
- $Z = 3552.825$

The Relative Centrifugal Force (RCF) for this tubular centrifuge is approximately **3553 g**.

Would you like to learn more about the structure and industrial applications of the tubular bowl centrifuge?

$$RCF = \frac{F_c}{F_g} = \frac{mr\omega^2}{mg} = \frac{r\omega^2}{g}$$

$$RCF = \frac{r\left(\frac{2\pi n}{60}\right)^2}{g}$$

$$RCF = \frac{r\left(\frac{2\pi n}{60}\right)^2}{9.8}$$

$$\boxed{RCF = 0.001119\, r n^2}$$

radius = 12.7 cm ; n = 5000 rpm

$$= 0.001119 \times (12.7) \times (5000)^2$$

$$= 3552.82$$

Question: Explain the terms “salting in” and “salting-out”. Emphasize
the role of Stern layer and provide information on the interactive forces involved in protein precipitation/salting out.
ii). Write the key principle involved in precipitating proteins with organic solvent highlighting its potential advantages and disadvantages.

Answer: **Salting-Out and Salting-In Salting-out** is a widely used precipitation method where high concentrations of salt (typically ammonium sulfate) are added to a protein solution. This increases the solution's ionic strength, which drastically decreases protein solubility and forces the proteins to aggregate and precipitate

- **The Role of the Stern Layer and Interactive Forces** In an aqueous solution at a near-neutral pH, proteins generally carry a net negative surface charge. This negative charge naturally attracts a tightly bound layer of positive counter-ions from the surrounding liquid, which is known as the **Stern layer**.
- Just outside the Stern layer is an "electrical double layer". When two similarly charged protein molecules approach one another, their double layers repel each other.
- Furthermore, water molecules bind to the hydrophilic surface of the protein to form a protective **hydration shell**. These repulsive electrostatic forces and the physical water barrier keep the proteins dissolved

proteins dissolved.

- However, proteins also inherently possess attractive forces, namely van der Waals forces and hydrophobic interactions.
- **During salting-out**, the massive influx of salt ions competes with the proteins for the available water molecules, effectively stripping away the protein's protective hydration shell.
- Once this water barrier is disrupted, the hidden hydrophobic zones on the protein's surface are exposed. The attractive hydrophobic interactions then overpower the repulsive electrostatic forces, causing the naked protein molecules to clump together and precipitate